



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON QUARTERLY EXAMINATION - 2024 - 25

Number

[Max. Marks : 70]

Time Allowed : 3.00 Hours]

PHYSICS

PART-I

Note: (i) Answer all the questions

15x1=15

(ii) Choose the most suitable answer from the given four alternatives and write the option code and the corresponding answer.

- One of the combinations from the fundamental Physical Constant is $\frac{hc}{G}$. The unit of this expression is
(a) Kg^2 (b) m^3 (c) S^{-1} (d) m
- If an object is dropped from the top of a building and it reaches the ground at $t = 4\text{s}$. Then the height of the building is (ignoring air resistance) ($g = 9.8 \text{ ms}^{-2}$)
(a) 77.3 m (b) 78.4 m (c) 80.5 m (d) 79.2 m
- The Centrifugal force appears to exist
(a) Only in Inertial Frames (b) Only in Rotating Frames
(c) In any accelerated Frames (d) Both in Inertial and Non - Inertial Frames
- When a car takes a sudden left turn in the curved road, passengers are pushed towards the right due to
(a) Inertia of direction (b) Inertia of Motion
(c) Inertia of Rest (d) Absence of Inertia
- What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?
(a) $\sqrt{2gR}$ (b) $\sqrt{3gR}$ (c) $\sqrt{5gR}$ (d) $\sqrt{gR}$
- A spring of force constant K is cut into two pieces such that one piece is double the length of the other. Then, the long piece will have a force constant of
(a) $\frac{2}{3} K$ (b) $\frac{3}{2} K$ (c) $3 K$ (d) $6 K$
- A couple produces,
(a) Pure Rotation (b) Pure Translation
(c) Rotation and Translation (d) No motion
- The speed of a solid sphere after rolling down from rest without sliding on an inclined plane of vertical height h is,
(a) $\sqrt{\frac{4}{3}gh}$ (b) $\sqrt{\frac{10}{7}gh}$ (c) $\sqrt{2gh}$ (d) $\sqrt{\frac{1}{2}gh}$
- If the mass and radius of the earth are both doubled. then the acceleration due to gravity 'g'
(a) remains same (b) $g/2$ (c) $2g$ (d) $4g$
- In an experiment, four quantities a , b , c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. Quantity P is calculated as follows. $P = \frac{a^3 b^2}{cd}$. % error in P is
(a) 7% (b) 4% (c) 14% (d) 10%
- Motion of a particle is given by equation $S = (3t^3 + 7t^2 + 14t + 8)\text{m}$. The value of acceleration of the particle at $t = 1\text{s}$ is
(a) 10 m/s^2 (b) 16 m/s^2 (c) 23 m/s^2 (d) 32 m/s^2
- Tension acting on the mango of mass 500g hanging from a tree is
(a) 3.92 N (b) 4 N (c) 4.9 N (d) 4.9 N
- Two bodies of masses m and $4m$ are moving with equal kinetic energies. The ratio of their Linear Momentum is
(a) $4 : 1$ (b) $1 : 4$ (c) $1 : 2$ (d) $2 : 1$
- A Force $\vec{F} = \alpha \hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about the origin is conserved is
(a) -1 (b) 1 (c) Zero (d) 2



15. A body weighs 200 N on the surface of the earth. How much will it weight held way down to the centre of the earth? (a) 250 N (b) 200 N (c) 100 N (d) 150 N

PART - II

Note : Answer any six questions. Question Number : 24 is Compulsory.

6x2=12

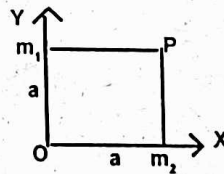
16. Define Precision and Accuracy.
17. A Radar signal is beamed towards a planet and its echo is received 7 minutes later. If the distance between the Planet and the Earth is 6.3×10^{10} m Calculate the speed of the signal?
18. Define Acceleration. Give its unit.
19. What are the steps followed for developing the free body diagram?
20. Consider a circular road of radius 20 meter banked at an angle of 15 degree. With what speed a car has to move on the turn so that it will have safe turn?
21. Write the difference between Conservative Force and Non - Conservative Forces. (any two)
22. Define Torque and Mention its Unit.
23. State Newton's Universal Law of Gravitation.
24. An object is thrown with initial speed 5ms^{-1} with an angle of projection 30. What is the height reached by the particle?

PART - III

Note : Answer any six questions. Question Number : 33 is Compulsory.

6x3=18

25. Write a note on Triangulation method and Radar method to measure larger distances.
26. Derive the equations of Motion for a Particle (a) Falling Vertically (b) Projected Vertically
27. Write the difference between Static Friction and Kinetic Friction.
28. Explain Work done by the constant force with Graph.
29. A bullet of mass 50g is fired from below into a suspended object of mass 450g. The object rises through a height of 1.8m with bullet remaining inside the object. Find the speed of the bullet. (Take $g = 10\text{ms}^{-2}$)
30. State and Prove Perpendicular Axis Theorem.
31. Find the moment of Inertia of a disc of mass 3Kg. and radius 50 cm about the following axes.
(i) Axis passing through the centre and Perpendicular to the plane of the disc.
(ii) Axis touching the edge and perpendicular to the plane of the disc.
32. State Kepler's three laws.
33. Two particles of masses m_1 and m_2 are placed along the the X and Y axis respectively at a distance 'a' from the origin. Calculate the Gravitational field at a point P shown in figure below.



PART - IV

Note : Answer all the questions.

5x5=25

34. (a) Obtain an expression for the time period T of a Simple Pendulum. The time period T depends on i) mass 'm' of the bob (ii) Length 'l' of the pendulum and (iii) Acceleration due to gravity g at the place where the pendulum is suspended. (Constant $K = 2\pi$)
(OR)
(b) Define Power. Derive the relation between Power and Velocity.
35. (a) Derive the equation of motion, Range and maximum height reached by the particle thrown at an Oblique angle θ with respect to the horizontal direction.
(OR)
(b) Explain in detail the Triangle law of Addition.
36. (a) Explain the motion of blocks connected by a string in Horizontal motion..
(OR)
(b) Derive an expression for acceleration and Speed of the particle when it reached the bottom in an inclined plane.
37. (a) Derive the expression for Moment of Inertia of a rod about its centre and Perpendicular to the rod.
(OR)
(b) Discuss rolling on inclined plane and arrive at the expression for the acceleration.
38. (a) Derive an expression for Escape Speed.
(OR)
(b) Derive an expression for energy of an Orbiting Satellite.